



Lean Agentic AI on Google Cloud

Engineering agents for energy, carbon, and cost.



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Before we deploy a thousand agents...

What does it actually cost to run an agent at production scale — in dollars, in watts, and in carbon?



SCALE IN 2026

The agentic gold rush.

TOKENS / MONTH

6T₊

Processed monthly by Gemini through the ADK.

DATA-CENTRE ELECTRICITY

+128%

From 415 TWh in 2024 to 945 TWh by 2030.

LONG-RUNNING OPS

7 days

Now supported by Agent Runtime — agents that don't sleep.

Enterprises are deploying agents at unprecedented speed — but the bill is only just starting to land.

The hidden bill of agentic AI.



COST

100x₊

per agent session

A coding-agent session can consume
~138x the energy of a single chat query —
costs compound proportionally.



ENERGY

3–10x

vs a web search

An LLM query consumes roughly 3 to 10
times the electricity of a traditional Google
search.



CARBON

+128%

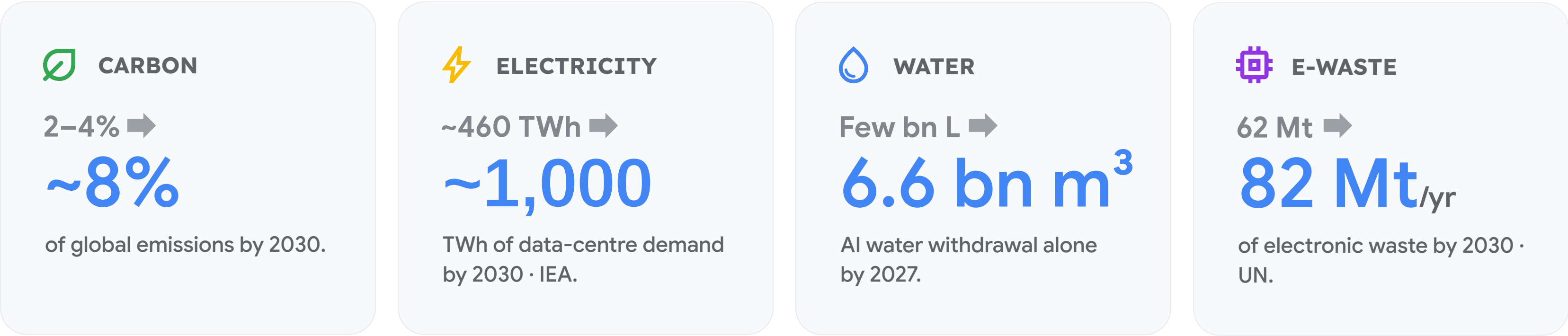
data-centre electricity

Global data-centre electricity demand is
set to more than double from 415 TWh
(2024) to 945 TWh by 2030.

These compound simultaneously — **every wasted token costs money, watts, and emissions at once.**

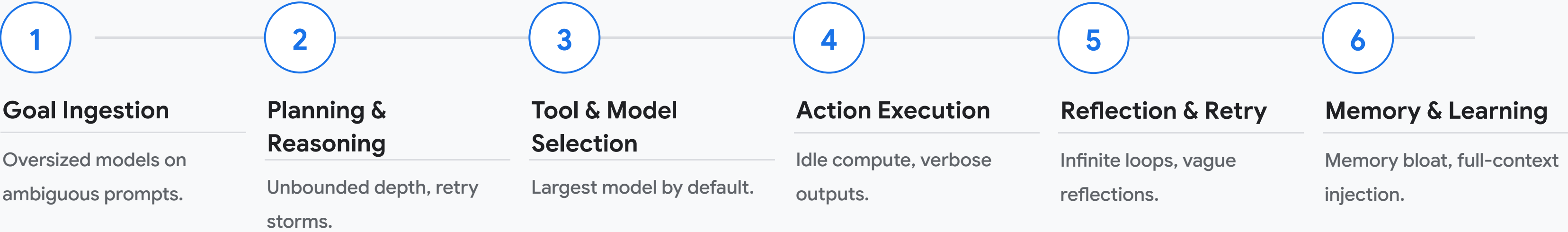
It's broader than cost and carbon.

Innovation gave us everything — but its footprint grew unseen, unattended, and unmeasured .



Different impacts. Different units. But the same story — every curve is going up.

Where the waste actually lives.



Every stage is a leak.

Waste at one stage compounds into the next — verbose plans drive longer tool calls, which drive larger memories, which drive bigger prompts on the next run.

The measurement problem.

Most teams reason about agent cost qualitatively.

“This pattern feels lean.”

“That model is probably cheaper.”

“We’ll optimize later.”

But cost, energy, and carbon are not intuitions. They are numbers — measurable, attributable, comparable. Without a measurement layer, every architectural choice is a guess and every “optimization” is unfalsifiable.

WHAT WE NEED



A standard unit

for the carbon impact of an agent task.



A measurement method

that uses telemetry we already collect.



An open standard

so results are comparable across teams, clouds, and vendors.

SCI for AI — the formula.

$$\text{SCI} = (E \times I) + M \text{ per } R$$

E ENERGY

consumed by the software, in kWh.

I INTENSITY

carbon per kWh of the grid powering it.

M EMBODIED

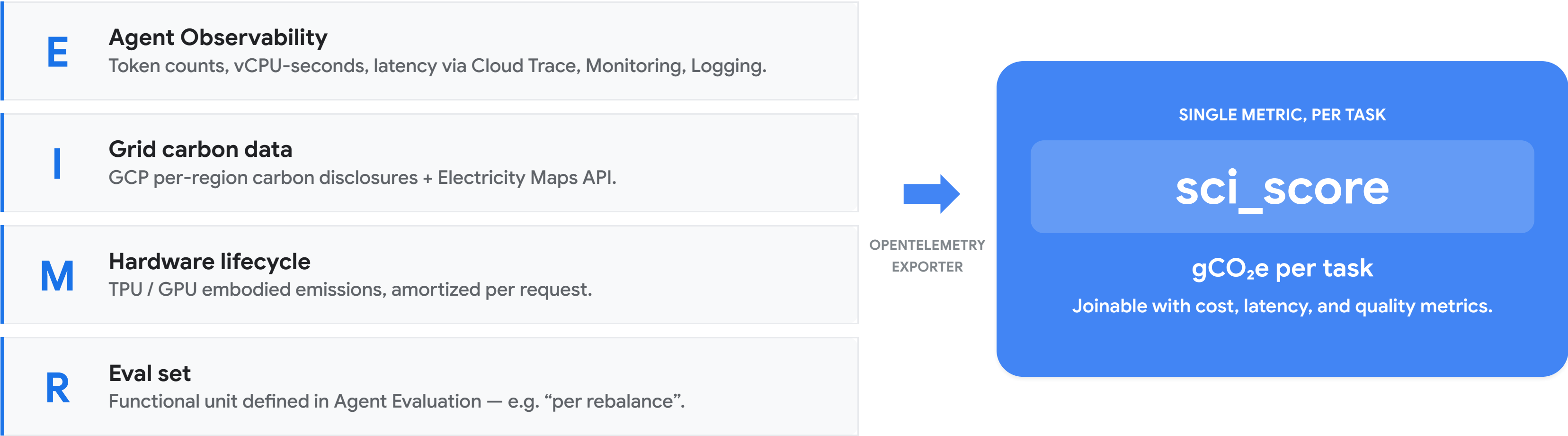
amortized hardware manufacturing carbon.

R FUNCTIONAL UNIT

the unit of work — e.g., per transaction, per prompt.

 Action-oriented: SCI talks about **reduction**, not offset.

Wiring SCI into Agent Observability.

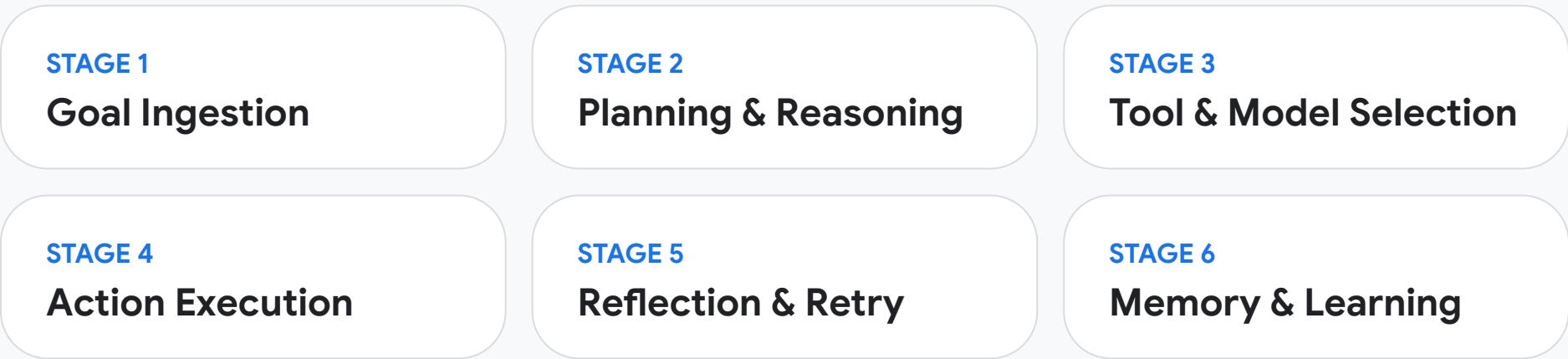


This is the spine. **Everything that follows is how to drive this number down.**

A DISCIPLINE, NOT A CHECKLIST

Introducing Lean Agentic AI.

A discipline for designing agents that are **efficient by design** — across cost, energy, and carbon.



FROM THE BOOK

Lean Agentic AI

Minimizing Cost, Carbon, and Complexity.

LEARN MORE leanagenticaai.com

THE RECURRING DESIGN CHOICES THAT MOVE THE SCI SCORE

The lean levers.



Model Selection

Smallest model that meets the bar.



Prompt Design

Structured, templated, minimal.



Context Window

Only what's needed, never more.



Tool Usage

Conditional, metadata-driven.



Reflection Caps

Bounded retries, calibrated confidence.



Memory Storage

High-signal only, TTL-based.



Execution Timing

Carbon-aware scheduling and regions.



Observability

Measure everything, attribute it.

BUILD, SCALE, GOVERN, OPTIMIZE

Gemini Enterprise Agent Platform.

BUILD	SCALE	GOVERN	OPTIMIZE
• Agent Studio • ADK (graph framework) • Model Garden • RAG Engine	• Agent Runtime (serverless) • Memory Bank + Sessions • Memory Profiles • Batch & Event agents	• Agent Registry • Agent Gateway • Agent Identity • Model Armor	• Agent Evaluation • Agent Simulation • Agent Observability • Example Store

Every lean lever maps to specific services across these four pillars.

The worked example.

“
Rebalance ESG portfolio for
client 84721.

Behind this single sentence, an agent must:

- Interpret intent, extract client ID, scope (equity / bonds / hybrid).
- Plan: data fetch → ESG scoring → risk → allocation → summary.
- Select scoring models, ESG filters, risk tools dynamically via MCP.
- Execute rebalancing logic and generate recommendations.
- Reflect on confidence, retry where constraints are unmet.
- Store accepted plans, client preferences, error history.

github.com/navveenb/lean-agentic-ai/tree/main/lean-agentic-google-adk-demo



DOMAIN

ESG portfolio rebalancing

Financial services · multi-tool · multi-model.

WHAT'S NEXT

We will walk this example through all **six stages** — default vs lean — on Google Cloud.

1

2

3

4

5

6

Stage 1 — Goal Ingestion.

DEFAULT

Send every prompt straight to Gemini Pro by default.
Inject full client history and past meetings as context.
Free-form output parsing with retries on malformed JSON.

LEAN

Two-stage funnel: Flash-Lite as intent classifier first, route to Pro only when needed.
Scoped retrieval from Memory Bank — last rebalance summary + ESG preferences only.
Structured output via `response_schema` — no retries on parsing.

KEY GCP SERVICES

Gemini 3.1 Flash-Lite · Gemini 3.1 Pro · Memory Bank (scoped retrieval) · ADK structured outputs

Stage 2 — Planning & Reasoning.

DEFAULT

Ad-hoc sub-agent spawning — full conversation passed between agents.

Unbounded planning depth, retry loops with near-identical prompts.

Pro used for every planning step, regardless of complexity.

LEAN

ADK graph framework with explicit DAG topology and bounded `max_depth`.

Scoped sub-agent state — pass only what each node

needs. Flash for orchestration logic, Pro reserved for final synthesis.

KEY GCP SERVICES

ADK graph-based framework · Conditional edges & scoped state · Gemini 3.1 Flash for planning · Pro for synthesis

Stage 3 — Tool & Model Selection.

DEFAULT

Largest model by default for every sub-task — scoring, filtering, summarizing.

Multiple ESG scorers called in parallel for the same asset class.

Tools hardcoded in agent logic; no fallback hierarchy.

LEAN

Conditional tool invocation via Agent Registry metadata (cost, latency, carbon tags).

Right-sized models from Model Garden — classifier, scorer, summarizer all different.

Managed MCP Servers for BigQuery, Cloud SQL, Spanner — no bespoke data plumbing.

KEY GCP SERVICES

Agent Registry · Managed MCP Servers (BigQuery, Cloud SQL, Spanner) · Model Garden · Gemma 4 / Flash-Lite

Stage 4 — Action Execution.

DEFAULT

Always-on GCE / GKE instances with idle compute waste.
Full ESG policy document resent with every LLM call.
2,000-token narrative reports generated unnecessarily.

LEAN

Agent Runtime serverless mode — sub-second cold start, scales to zero.
Gemini context caching for reused policy and instruction blocks.
Gemini Batch API for non-real-time rebalancing — ~50% cost reduction .

KEY GCP SERVICES

Agent Runtime (serverless) · Gemini context caching · Gemini Batch API · Structured outputs via response_schema

Stage 5 — Reflection & Retry.

DEFAULT

Re-prompt Pro 4–5 times with “try again, make it clearer”.
Resend full client history and prior outputs on every retry.
No calibrated confidence threshold — retries based on intuition.

LEAN

Agent Evaluation defines confidence thresholds against an eval set.
Example Store reuses known-good few-shot patterns — no LLM call needed.
Flash-Lite as a lightweight verifier; ADK node-level retry caps.

KEY GCP SERVICES

Agent Evaluation · Agent Simulation · Example Store · ADK retry policies · Flash-Lite verifiers

Stage 6 — Memory & Learning.

DEFAULT

Store every output, prompt, and tool response — unstructured blobs.

Inject full 3-session memory into every prompt.

Failed plans kept in recall set, polluting future inference.

LEAN

Memory Bank with event-triggered generation — only on [plan_accepted](#) events.

Memory Profiles separate persistent preferences from session scratch state.

Retention policies (30-day TTL) and surgical semantic retrieval via [top_k](#).

KEY GCP SERVICES

Agent Platform Memory Bank · Memory Profiles · Agent Platform Sessions · Event-triggered generation

RUN THE RIGHT WORK IN THE RIGHT REGION AT THE RIGHT TIME

Carbon-aware scheduling on GCP.



LEVER 1 • REGION

Region selection

Place batch agent workloads in regions with high carbon-free energy percentages (Google publishes this per region).



LEVER 2 • TIME

Deferred batch queues

Use Pub/Sub + Batch agents to defer non-urgent sub-tasks. Trigger on low grid-intensity windows via Electricity Maps.



LEVER 3 • TRIAGE

Real-time stays put

Interactive agents can't be moved — but you can split urgent vs deferred sub-tasks and route accordingly.

KEY GCP SERVICES

GCP per-region carbon disclosures • Batch & Event-driven agents • Pub/Sub triggers • Electricity Maps integration

A PEOPLE CHANGE FIRST, A SYSTEMS CHANGE SECOND

The cultural shift.



01 PEOPLE

Executive Advocacy

Cost-and-carbon must be a stated leadership priority, not a side-bar.



02 METRICS

KPIs for Cost + Carbon

SCI per task tracked alongside latency and quality on every agent.



03 MINDSET

Mindset Shift

Engineering teams reward lean design as much as feature velocity.



04 ENABLEMENT

Education & Enablement

Teams need training on what “lean” actually means in each stage.



05 ORG

Cross-Disciplinary

Sustainability, FinOps, and engineering teams share a single dashboard.



06 DISCIPLINE

Continuous Optimization

SCI scores reviewed quarterly; regressions treated like performance bugs.

Your Monday-morning checklist.

- 1 Route through [Flash-Lite first](#) — invoke Pro only when confidence demands it.
- 2 Define your ADK graph topology explicitly — set `max_depth`, scoped sub-agent state.
- 3 Deploy on [Agent Runtime serverless](#) — scale-to-zero, sub-second cold starts.
- 4 Turn on [Gemini context caching](#) for any reused policy or instruction document.
- 5 Move overnight or batch jobs to the [Gemini Batch API](#) for ~50% cost reduction.
- 6 Configure [Memory Bank](#) with event-triggered generation — store only what was accepted.
- 7 Set ADK retry caps and confidence thresholds calibrated via [Agent Evaluation](#).
- 8 Enable [Agent Observability](#) and emit an `sci_score` metric per task.
- 9 Place batch workloads in [high-CFE regions](#) ; defer non-urgent work to low-carbon windows.

RESULT

A baseline `sci_score`, a measurable trajectory, and architecture you can defend in front of a CFO.

THE ONE IDEA TO TAKE HOME

You wouldn't send a Ferrari
to deliver a pizza ... 

Sure, it *could*. But would you?

Think Big. Design and Deploy Lean.

Thank you

Build agents that earn their watts.

BOOK

Lean Agentic AI

leanagenticai.com

STANDARD

SCI for AI

sci.greensoftware.foundation

PLATFORM

Gemini Enterprise Agent Platform

cloud.google.com

COMMUNITY

Green Software Foundation

greensoftware.foundation



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References & sources.

PLATFORM

Google Cloud — Introducing Gemini Enterprise Agent Platform

Cloud Next '26 announcement, April 2026.

cloud.google.com/blog/products/ai-machine-learning/introducing-gemini-enterprise-agent-platform

ENERGY & CARBON

IEA — Energy and AI (*special report*)

International Energy Agency, April 2025. The most comprehensive global analysis on AI's energy demand.

iea.org/reports/energy-and-ai

STANDARD

Software Carbon Intensity (SCI) Specification

Green Software Foundation; ISO/IEC 21031:2024. SCI for AI extension under development.

sci.greensoftware.foundation

INFERENCE ENERGY

Energy footprint comparisons — multiple studies

de Vries (Joule, 2023); OpenAI disclosure (2025); Couch / cnaught.com on agent sessions (Jan 2026). Estimates vary by methodology.

BOOK · COMPANION TO THIS TALK

Lean Agentic AI: Minimizing Cost, Carbon, and Complexity — Navveen Balani

LEARN MORE

leanagenticalai.com